

What is Samba?

Samba is a software package that allows a Linux system to **share files and printers** with **Windows, Linux, and macOS** computers over a network using the **SMB (Server Message Block)** protocol

What is SMB?

SMB = Server Message Block

It is a network protocol used for:

- File sharing
- Printer sharing
- Folder sharing

Windows uses SMB by default.

Why do we use Samba?

Linux

|

Share Folder

|

Windows PC

Package: Samba

Port number: 139,445 and 137,138

Configuration File : /etc/samba/smb.conf

Services: smb.service

Make the dir and inside create the file using the command touch

```
[root@client ~]# mkdir /birds
[root@client ~]# cd /
[root@client /]# cd birds/
[root@client birds]# ls
```

```
[root@client birds]# touch file{1..5}
[root@client birds]# ls
file1 file2 file3 file4 file5
```

Create Samba user and set smb password

```
#useradd samba
```

```
#smbpasswd -a sambauser
```

```
[root@client ~]# useradd samba
```

```
[root@client ~]# smbpasswd -a samba
New SMB password:
Retype new SMB password:
Added user samba.
```

Install the Package

```
#yum install samba
```

```
[root@client ~]# yum install samba*
CentOS Stream 9 - BaseOS                               5.3 kB/s | 5.4 kB    00:01
CentOS Stream 9 - AppStream                             5.8 kB/s | 5.5 kB    00:00
CentOS Stream 9 - Extras packages                       7.4 kB/s | 5.9 kB    00:00
Package samba-client-libs-4.23.5-8.el9.x86_64 is already installed.
Package samba-common-4.23.5-8.el9.noarch is already installed.
Package samba-common-libs-4.23.5-8.el9.x86_64 is already installed.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
samba                         x86_64            4.23.5-8.el9      baseos             869
samba-client                  x86_64            4.23.5-8.el9      appstream          753
samba-common-tools            x86_64            4.23.5-8.el9      baseos             494
samba-dc-libs                 x86_64            4.23.5-8.el9      baseos             117
samba-dcerpc                  x86_64            4.23.5-8.el9      baseos             722
samba-gpupdate                noarch            4.23.5-8.el9      appstream           17
=====
```

Modify /etc/samba/smb.conf file

```
#vi /etc/samba/smb.conf file
```

```
[root@client ~]# vi /etc/samba/smb.conf
```

Firewall disable

```
[root@client ~]# systemctl restart smb.service
^[[A[root@client firewall-cmd --permanent --add-service=samba
success
[root@client ~]# firewall-cmd --reload
success
```

Or

```
#systemctl disable firewalld.service
```

```
[root@client ~]# setenforce 0
[root@client ~]# getenforce
Permissive
```

start and enable the service

#systemctl enable --now samba

```
[root@client ~]# systemctl restart smb.service
```

```
[root@client ~]# systemctl enable --now smb.service
```

Verify the conf file

```
[root@client ~]# systemctl enable --now smb.service
[root@client ~]# testparm
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_STANDALONE

Press enter to see a dump of your service definitions
```

Ensure that the directory has been shared or not

#smbclient -L \\localhost

```
[root@client ~]# smbclient -L //localhost
Password for [SAMBA\root]:
Anonymous login successful

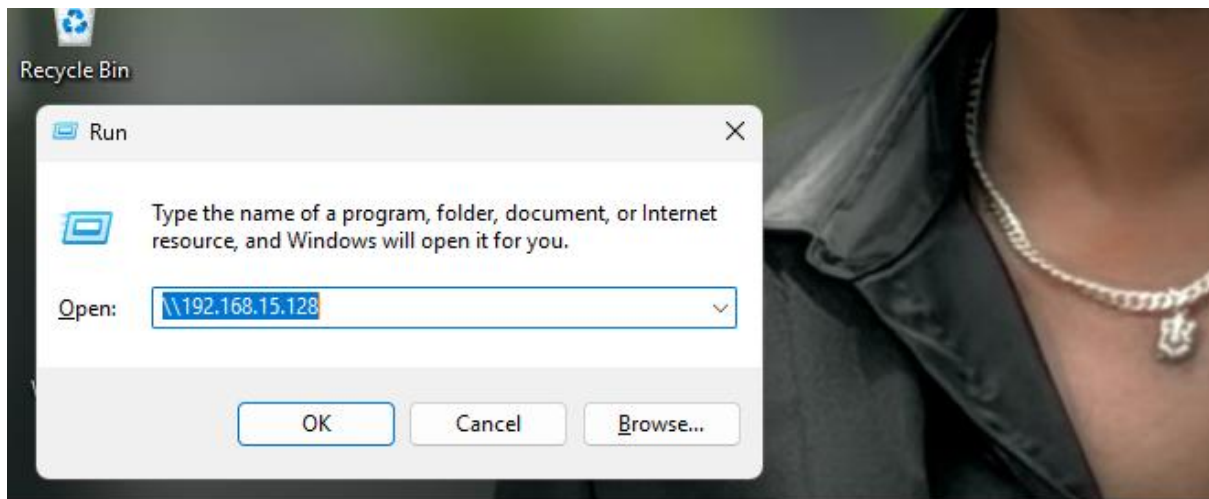
      Sharename      Type      Comment
      -----
      print$         Disk      Printer Drivers
      samba          Disk      sharing dir to window
      IPC$           IPC       IPC Service (Samba 4.23.5)
```

Check ip a

```
[root@client ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b4:07:9c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.15.128/24 brd 192.168.15.255 scope global dynamic noprefixroute ens33
        valid_lft 1028sec preferred_lft 1028sec
    inet6 fe80::20c:29ff:feb4:79c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

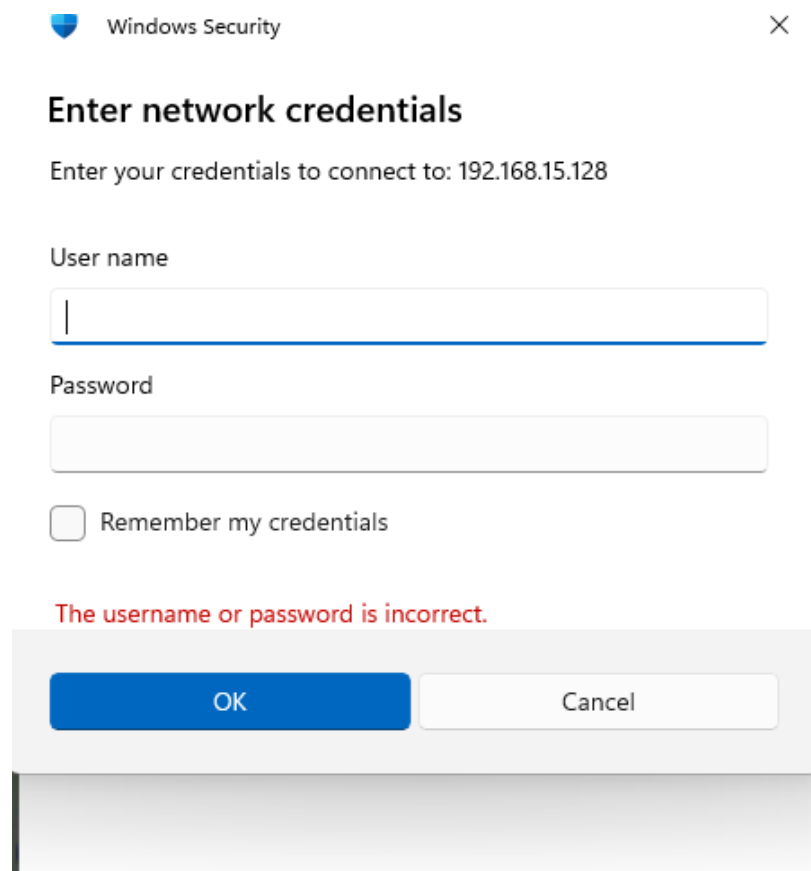
In Windows

Press Win + r

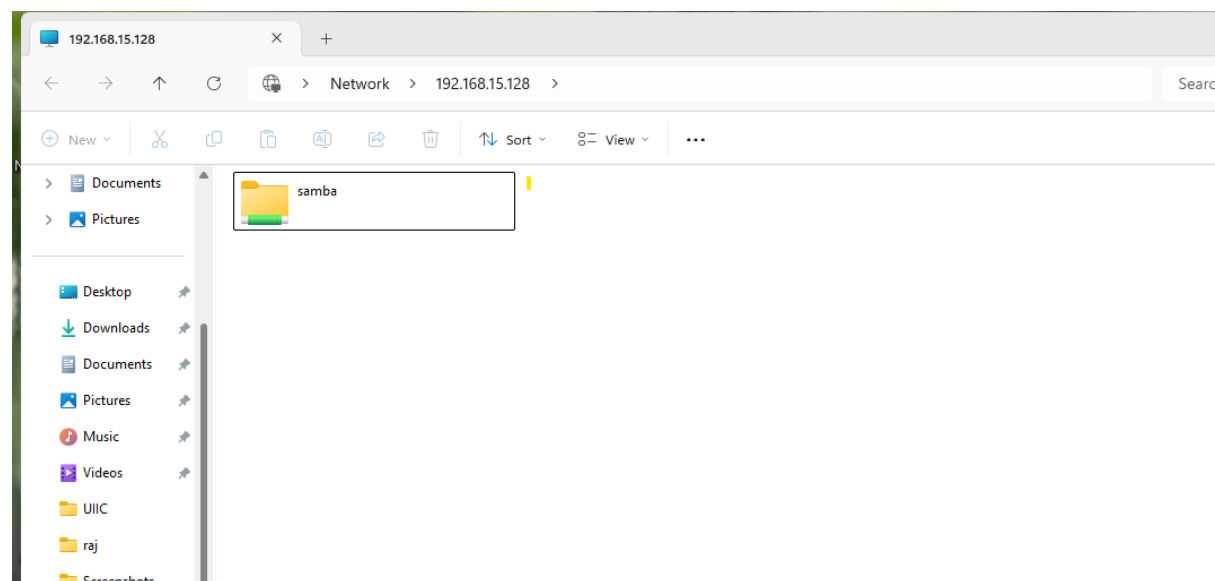


Enter username and password

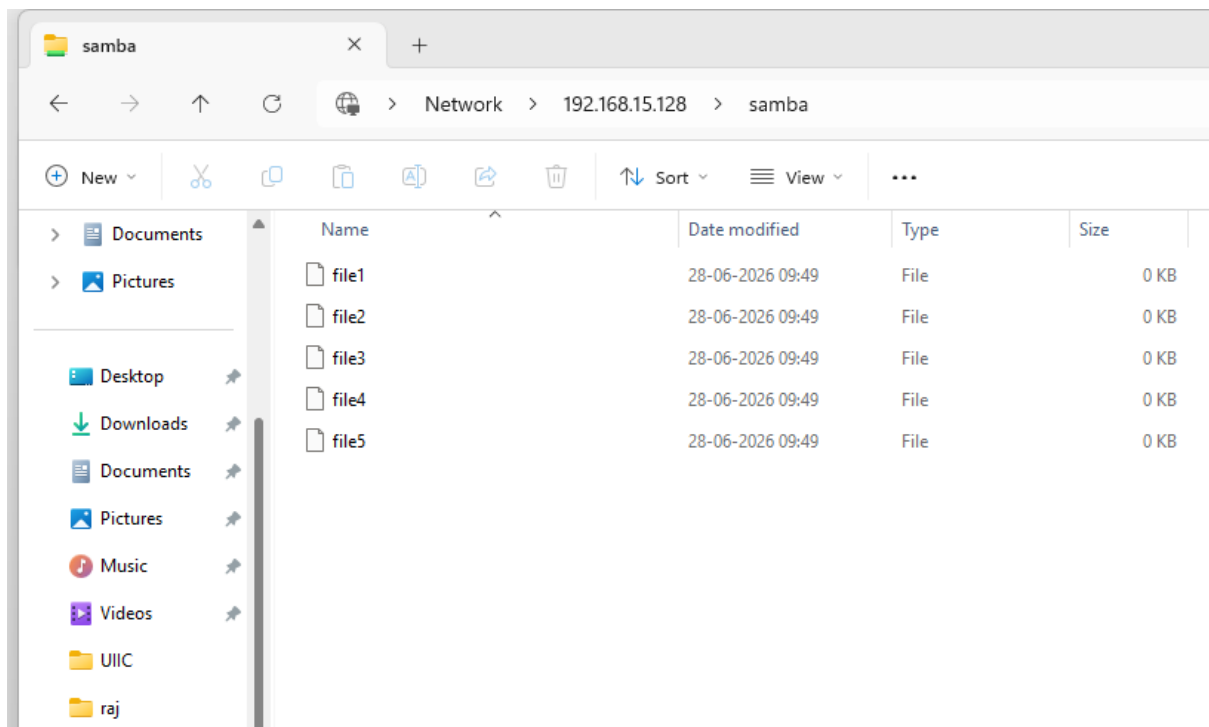
To access the file



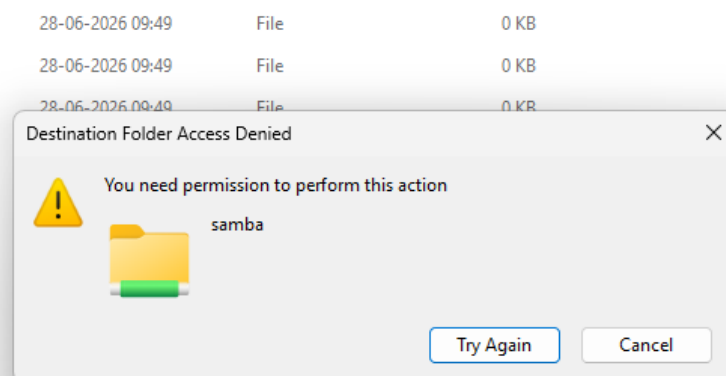
Open the file



Show the file



Bur I will not create the folder



So I give the full permission

chmod

- Change permissions.

777

- Everyone can:
 - Read
 - Write
 - Execute

```
valid_lft forever preferred_lft f  
[root@client ~]# chmod 777 /birds
```

Then I will create the new folder

Desktop	✦	file3	28-06-2026 09:49	File	0 KB
Downloads	✦	file4	28-06-2026 09:49	File	0 KB
		file5	28-06-2026 09:49	File	0 KB
Documents	✦	New folder	28-06-2026 10:17	File folder	
Pictures	✦				
Music	✦				